Investigating functional regulators of cold-induced pausing in mouse embryonic stem cells

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Summary

Embryonic development can be reversibly paused in response to mild hypothermia, yet the molecular regulator contributing to this process is unknown[1]. Establishing upon preliminary proteomics data identifying two cold-inducible RNA-binding proteins CIRBP and RBM3 as significantly upregulated at 30°C, this study investigated whether *Cirbp* and *Rbm3* are essential to sustain proliferation and viability of mouse embryonic stem cells (mESCs) during cold-induced pausing. siRNA-mediated knockdown revealed that *Rbm3* depletion led to a modest reduction in cell proliferation at 30°C, whereas *Cirbp* knockdown showed minimal effect. Additionally, culturing knockout clones of *Nodal*, a key regulator of metabolic reprogramming in the mTOR inhibition-induced diapause, generated an impaired cell proliferation at 30°C, implicating *Nodal* may be pivotal in both pausing models. The present studies support further investigation of *Rbm3* using complete knockout models to more precisely assess its effect on cell proliferation, and potentially define its associated mechanisms underlying cold-induced pausing in mESCs.

Introduction

Early mammalian embryonic development involves a sequential process of cleavage, compaction, and the formation of a blastocyst capable of implantation[2]. This progression is coordinated and driven by rigorous spatiotemporal interactions between the embryo and the maternal environment through hormonal signaling, nutrient sensing, and local cellular crosstalk, ensuring that development proceeds in synchrony with uterine receptivity[1,3]. Despite its programmed continuity, early embryonic development also exhibits a degree of plasticity[2]. Under adverse environmental or physiological stress, embryos can transiently suspend animation at the pre-implantation blastocyst stage[1,2]. This reversible development arrest, termed embryonic diapause, has been documented in over 130 mammalian species as an adaptive response to diverse environmental stressors such as seasonal photoperiods, nutrient limitation, concurrent lactation, and temperature fluctuations[1,4]. By enabling organisms to temporally uncouple fertilization from implantation, diapause provides a biological framework for integrating external cues into the temporal regulation of embryonic development. During

this state, embryos undergo a global reduction in cellular proliferation, transcriptional and translational activity, and metabolic rate, while their pluripotency and developmental potential remain intact[1,3-5]. The restoration of favorable conditions triggers dormant blastocysts for reactivation, resuming development toward implantation[6,7].

In mice, embryonic diapause occurs naturally during lactation, where the suckling stimulus triggers a hormonal milieu that prevents the estrogen surge required for implantation[1,3]. This state can also be experimentally recapitulated *in vivo* through surgical interventions such as ovariectomy, or via the injection of progesterone to inhibit estrogen signaling, representing a controlled switch governed by hormonal signals[4-6]. At the molecular level, a central pathway implicated in diapause regulation is the mechanistic target of rapamycin (mTOR) signaling pathway, which functions as a master nutrient and energy sensor to orchestrate anabolism and cellular growth[7,8]. While active mTOR signaling promotes high rates of translation and cell cycle progression, its inhibition leads to a shift toward catabolism and energy conservation[6-8]. Previously, studies from my lab have demonstrated that pharmacological inhibition of mTOR (mTORi) using the catalytic inhibitor INK128 induces a diapause-like state in both mouse blastocysts and embryonic stem cells (mESCs)[8,9]. These cells recapitulate *in vivo* diapause, as evidenced by cell cycle arrest, reduced biosynthesis, and a metabolic shift from glycolysis toward fatty acid oxidation[6,8,9]. Furthermore, transcriptomic profiling has confirmed strong similarity between mTOR-inhibited mESCs and the epiblast of diapaused blastocysts, supporting the validity of this *in vitro* model[9]. Within this framework, a member of the TGF- β superfamily, *Nodal*, was identified as a critical factor for mESCs survival and metabolic reprogramming under mTORi-induced diapause[10]. It was shown to enforce lipid catabolism by repressing peroxisome proliferator-activated receptor gamma (*Pparg*), maintaining cellular viability survival through active regulation of energy metabolism[10]. Given that global metabolic rewiring is a defining feature of diapause(-like) stage across both *in vivo* and *in vitro* contexts, *Nodal* may serve as a broader diapause regulator of metabolic adaptation extending beyond the mTORi model.

While studies across multiple mammalian cell types have demonstrated that mild hypothermia (28°C–32°C) can reversibly suppress cell cycle progression and induce growth arrest into dormancy, its underlying regulatory mechanisms and key regulators remain not fully uncharacterized[11,12]. Preliminary work from my lab has demonstrated that culturing mESCs under mild hypothermic conditions (30°C) induces a reversible pausing phenotype

characterized by reduced proliferation and translational activity, as well as a lower RNA content per cell. These features resemble those observed in the mTORi-induced pausing, suggesting that cold stress may represent an alternative environmental trigger of embryonic pausing *in vitro*. To identify molecular regulators of this cold-induced paused state, proteomic analysis comparing mESCs cultured at 30°C and 37°C revealed significant upregulation of two cold-shock RNA-binding proteins: Cold-Inducible RNA-Binding Protein (CIRBP) and RNA-Binding Motif Protein 3 (RBM3). CIRBP and RBM3 belong to a conserved family of glycine-rich RNA-binding proteins containing an N-terminal RNA recognition motif (RRM) and a C-terminal arginine-glycine-rich (RGG) domain[13]. The expression of *Cirbp* and *Rbm3* is selectively induced under mild hypothermic conditions (32°C–34°C) through temperature-dependent alternative splicing. For *Rbm3*, cold exposure promotes exclusion of a “poison” exon containing premature stop codons, thereby preventing the targeting nonsense-mediated decay and enabling stable mRNA production for rapid protein accumulations. In contrast, *Cirbp* is regulated through alternative 3’ end processing[13,14]. Functionally, CIRBP and RBM3 act as RNA chaperones that regulate gene expression through post-transcriptional modifications[13,14]. They preferentially bind to the 3-untranslated regions of target mRNAs, manipulating alternative polyadenylation (APA) to enhance transcript stability[13]. Beyond their shared molecular functions, CIRBP and RBM3 exhibit context- and cell-specific biological roles. For example, CIRBP is an essential to regulate the oscillation amplitude of core circadian genes in response to thermal fluctuations through APA regulation[13-15]. RBM3, in particular, has been extensively characterized within neuronal models, where it acts as a critical neuroprotector capable of preserving synaptic maintenance and regeneration, ensuring neuronal plasticity under neurodegenerative and traumatic brain injury conditions[16]. Additionally, the presence of *Rbm3* has enhanced resistance to apoptosis in human somatic cells cultured at 32°C[17]. Despite the established roles of *Cirbp* and *Rbm3* as stress-responsive regulators involved in many cellular physiological contexts, their functional contribution toward cold-induced pausing specifically in mESCs remains unexplored.

Therefore, I hypothesized that *Cirbp* and *Rbm3* encode essential regulators required to maintain mESC proliferative capacity and survival under cold-induced pausing. In parallel, given the role of *Nodal* in supporting metabolic adaptation during the mTORi-induced diapause, I aimed to determine whether *Nodal* functions as a universal survival factor that also sustains cell viability during cold-induced pausing, potentially through overlapping regulatory mechanisms. To test these hypotheses, I employed a combination of genetic perturbation and

functional assay to assess the functional necessity of these candidates. *Nodal* knockout mESC lines were grown under both control (37°C) and cold stress (30°C) conditions to assess the impact of *Nodal* deficiency on cell proliferation. In parallel, siRNA-mediated knockdown of *Cirbp* and *Rbm3* was performed to reveal their individual contributions to cell proliferation and viability during cold-induced pausing.

My current findings indicate that *Nodal* deficiency leads to an impairment on the proliferative capacity and potentially viability of mESCs under cold stress conditions. Furthermore, *Rbm3* depletion results in a modest reduction in cell proliferation compared to *Cirbp* knockdown, suggesting a more prominent role for *Rbm3* in maintaining the cold-paused state of mESCs. Taken together, these results provide a molecular basis for cold-induced mESC pausing, highlighting the involvement of two specialized cold-shock proteins, CIRBP and RBM3. Future investigation of *Rbm3* may yield insights into a key regulatory mechanism underlying cell survival or proliferative control during cold-induced pausing.

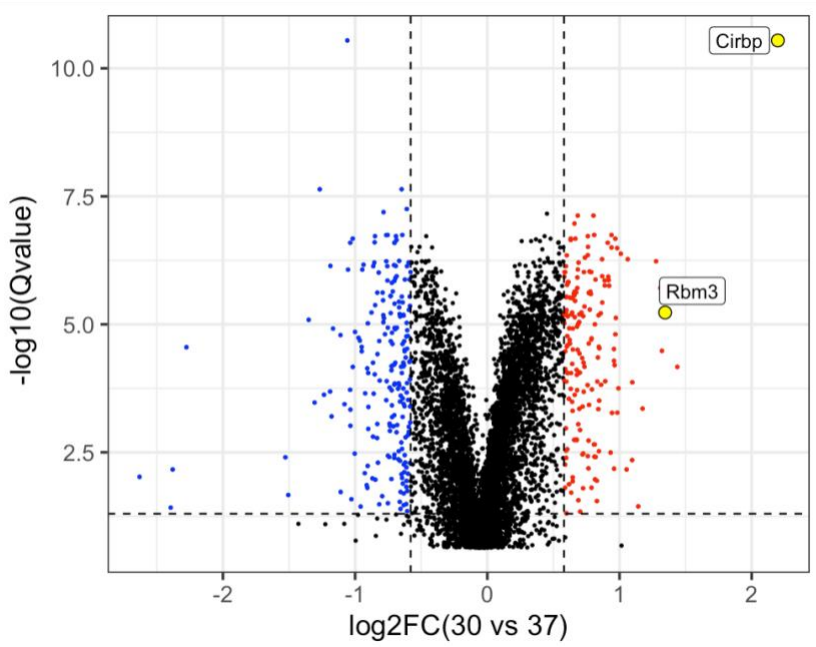


Figure 1. Differential protein expression in mESCs showing significant upregulation of CIRBP and RBM3 at 30°C. Volcano plot showing differential protein expression in mESCs grown in culture for 48 hours at 30°C compared to 37°C, identified by mass spectrometry. Significantly upregulated proteins (red) were determined using q-value < 0.05 and absolute Log2 fold change > 0.5. CIRBP and RBM3 are highlighted in yellow.

Results

1. *Nodal* deficiency shows a moderate impair during cold-induced pausing

To determine whether *Nodal* sustains a functional role in regulating cell proliferation under cold-paused conditions, wild-type E14 mESCs (WT) and two independent *Nodal* knockout ESC lines (KO1 and KO2) were cultured at 37°C and 30°C over four days. Cell numbers were normalized to the initial seedling density at D0 to control for plating variability.

Under control conditions (37°C), all cell lines exhibited a progressive proliferation during D0–D2, followed by a pronounced acceleration in growth from D3 onward. By D4, WT mESCs expanded to approximately 53-fold relative to the D0 baseline, whereas *Nodal* KO1 and KO2 ESCs reached ~28-fold and ~23-fold increases, respectively. Although both KO lines showed a roughly 50% reduction in proliferative compared to WT, their overall growth trajectories remained comparable throughout, indicating a consistent proliferative capacity between the KO lines. The reduction in normalized cell number of *Nodal* KO1 and KO2 became statistically significant by D4 ($p < 0.0001$), leading to a divergence in cell proliferation between KO lines and WT under the control condition (Figure. 2A).

Under cold-paused conditions (30°C), proliferation of all cell lines was markedly attenuated relative to the control condition, consistent with a defined pausing phenotype. WT mESCs showed a steady increase in normalized cell number, reaching ~2.7-fold expansion by D4. In contrast, *Nodal* KO1 and KO2 ESCs failed to sustain growth during later days of the experiment. The proliferation of *Nodal* KO1 ESCs plateaued between D3 and D4, while KO2 cells showed a modest decline over the same period, both remaining significantly lower than the WT ($p < 0.0001$). Notably, while *Nodal* deficiency seemed to reduce proliferation under both temperature conditions, the defect in cell proliferation was stronger under 30°C conditions. Comparing to the reduced yet sustained proliferation observed at 37°C, *Nodal* KO ESCs showed minimal expansion beyond baseline when cultured at 30°C (Figure. 2B).

Collectively, these data indicate that *Nodal* may support mESC proliferation and ensure basal cell viability under the cold-induced pausing state.

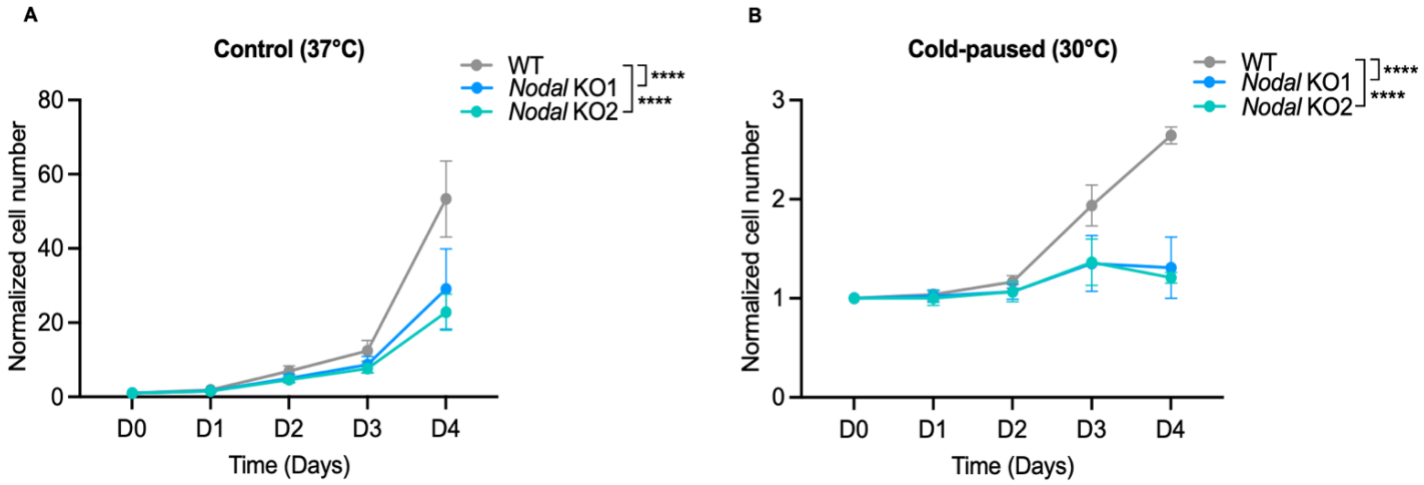


Figure 2. *Nodal* deficiency impairs mESC proliferation under cold stress conditions. Wild-type (WT) and two independent *Nodal* knockout mESC lines (KO1 and KO2) cultured at 37°C (left) and 30°C (right) over four days. Cell numbers were normalized to the initial seedling density at D0. Statistical significance between WT and each *Nodal* KO line was determined by two-way ANOVA; ****$p < 0.0001$. Data are presented as mean ± SD, n = 3 biological replicates.

2. Impacts of *Cirbp* and *Rbm3* depletion on cell growth under cold-induced pausing

To investigate whether *Cirbp* and *Rbm3* contribute to the regulation of cell survival and proliferation under cold-paused conditions, WT mESCs were transfected with 30 nM siRNA to selectively knockdown *Cirbp* (*Cirbp* KD), *Rbm3* (*Rbm3* KD), and in combination (*double* KD), coupling with non-targeting control siRNA (NTC) as reference conditions. The transfected mESCs were cultured at 37°C and 30°C over six days. Again, cell numbers were normalized to D0 to control for plating viability in the initial seedling density.

At 37°C, all knockdown conditions demonstrated a comparable proliferative expansion with NTC. From D0 to D4, cell growth has remained similar across all groups, followed by a robust proliferation from D5 onward. By D6, *Cirbp* KD cells reached the highest normalized cell count, while other KD lines and NTC converged by showing a modestly lower cell number that was not statistically significant ($p > 0.05$) (Figure. 3A). This indicates that *Cirbp* and *Rbm3* depletion does not substantially impact mESC proliferation under control conditions.

At 30°C, NTC cells exhibited a gradual increase in normalized cell number, reaching approximately 9-fold expansion by D6. However, a distinct proliferation pattern has emerged among the KD conditions. *Cirbp* KD cells exhibited a proliferation trajectory similar with NTC, with only a mild reduction in cell number on the last day that did not reach statistical significance. In contrast, the proliferation of *Rbm3* KD cells remained closely to that of double

KD, suggesting the observed proliferation attenuation in the double knockdown condition may be primarily driven by *Rbm3* loss (Figure. 3B).

Collectively, these data indicate that *Rbm3* depletion leads to progressive reduction in mESC proliferation during cold-induced pausing, whereas *Cirbp* knockdown exhibits a mostly neutral phenotype, supporting a distinct role of *Rbm3* mediating cellular adaptation in response to cold stress.

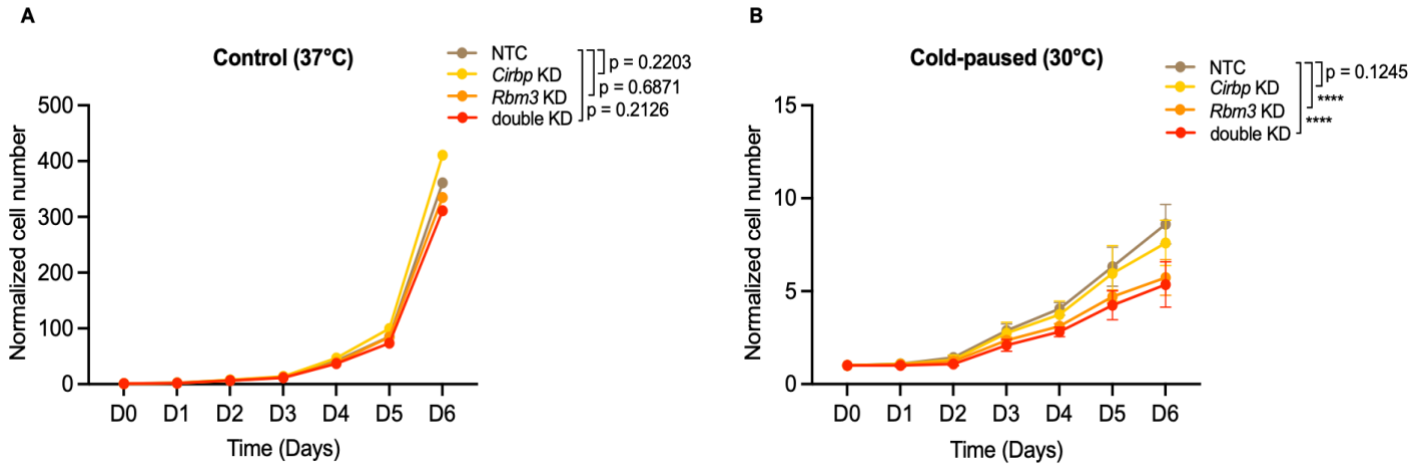


Figure 3. Effects of *Cirbp* and *Rbm3* knockdown on mESCs proliferation at 37°C versus 30°C. mESCs transfected with non-targeting control siRNA (NTC), siRNA targeting *Cirbp* (*Cirbp* KD), siRNA targeting *Rbm3* (*Rbm3* KD), and both (double KD) were cultured at 37°C (left) and 30°C (right) over six days. *Rbm3* KD and double KD ESCs showed significant reduction in expansion relative to NTC. Statistical significance between NTC and each knockdown condition was determined by two-way ANOVA; ****p < 0.0001. Data for 37°C are presented as mean, n = 2 biological replicates; data for 30°C are presented as mean ± SD, n = 3 biological replicates.

3. Efficient depletion of *Cirbp* and *Rbm3* at transcript and protein level

Following 48 hours and 72 hours post-transfection, qRT-PR and western blot were performed to assess the efficacy of siRNA-mediated knockdown of *Cirbp* and *Rbm3* at both transcript and protein levels.

At the transcript level, siRNA targeting *Cirbp* at resulted in a reduction of *Cirbp* mRNA expression to 0.69-fold (31% knockdown) at 37°C. The knockdown efficiency of *Cirbp* was enhanced to 0.40-fold (60% knockdown) at 30°C, while double knockdown groups showed equivalent levels of silencing for both gene (~60–70%), indicating that co-transfection did not compromise single gene knockdown efficiency (Figure. 4A).

Notably, under both 37°C and 30°C conditions, single-gene knockdown of either *Cirbp* or *Rbm3* has resulted in an overexpression of its non-targeted paralog. For example, siRNA *Cirbp* transfection increased *Rbm3* expression to 1.19-fold under 37°C and 1.29-fold under 30°C, respectively (Figure 4. A). This suggests a potential compensatory response between *Cirbp* and *Rbm3* at the transcript level, independent of temperature conditions.

Consistent with the qRT-PCR results, western blot analysis confirmed a corresponding protein reduction of CIRBP and RBM3 under their respective knockdown conditions. The protein intensity of CIRBP and RBM3 was reduced in both single and double knockdown samples relative to the NTC (Figure 4. B). The degree of protein depletion generally aligned with the observed percentage of knockdown reflected by mRNA expression. These results confirm that 30 nM siRNA of *Cirbp* and *Rbm3* at 30 nM is sufficient to mediate effective knockdown and lead to differential proliferative phenotypes, which further supports the validity of proliferation assay analysis.

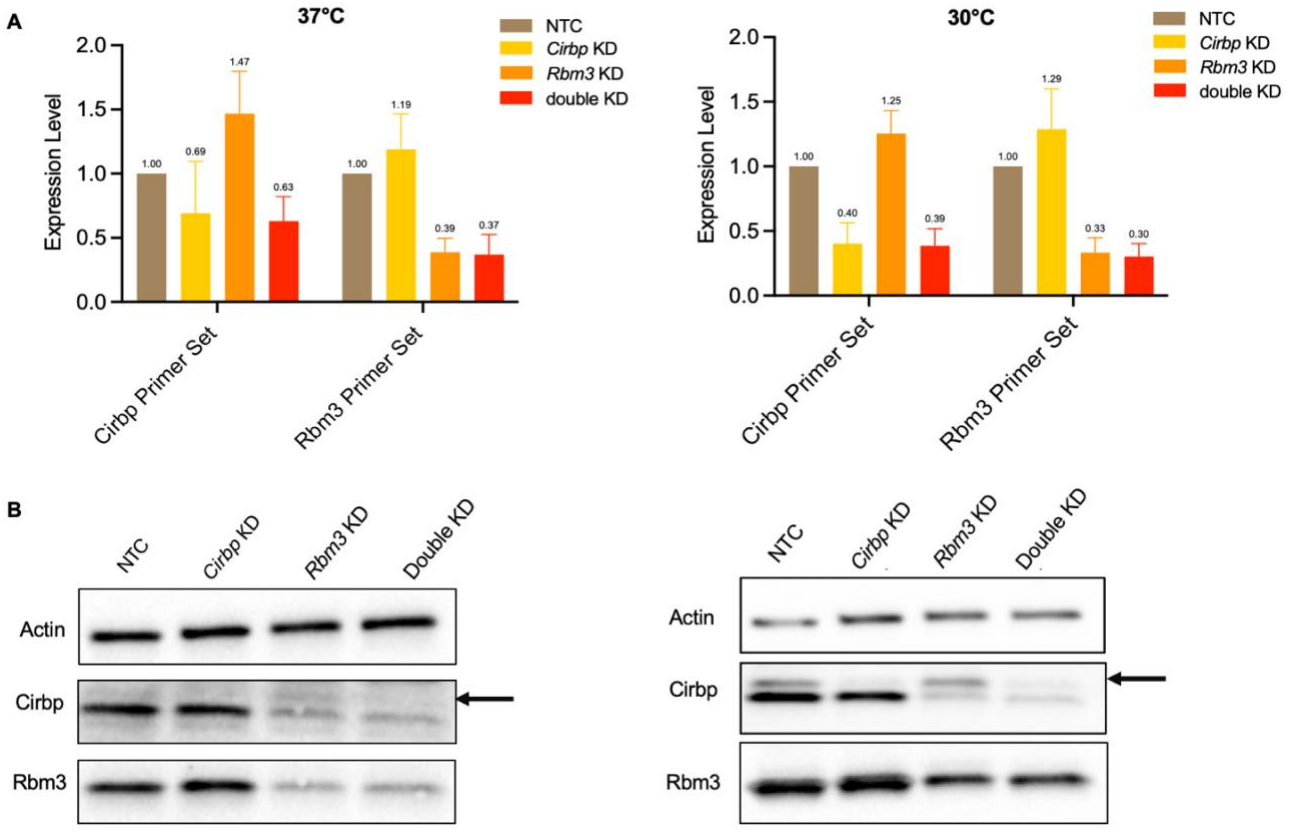


Figure 4. Efficient knockdown of *Cirbp* and *Rbm3* validated at transcript and protein levels. (A) Wild-type E14 mESCs were transfected with 30 nM non-targeting control siRNA (NTC), siRNA targeting *Cirbp* (*Cirbp* KD), siRNA targeting *Rbm3* (*Rbm3* KD), and both (double KD) at 37°C (left) and 30°C (right). Relative expression was measured by qRT-PCR using ΔΔCt method. The expression level of each knockdown condition was normalized to Ubb

housekeeping gene and presented relative to NTC. Data are presented as mean ± SD, n = 3 biological replicates (with 3–4 technical replicate per biological replicate). **(B)** Western blot analysis of CIRBP and RBM3 protein levels at 37°C (left) and 30°C (right), with Actin as loading control. The protein bands of CIRBP are indicated by arrows.

Discussion

In this study, I sought to characterize the functional contribution of *Nodal*, *Cirbp*, and *Rbm3* to the regulation of mESC proliferation and survival under cold-induced pausing. The results demonstrate that Nodal deficiency impairs mESC proliferative capacity under both control and cold-paused conditions, with a stronger phenotype seen under 30°C. Under the cold stress treatment, *Nodal* KO mESCs fail to sustain the proliferative expansion observed in WT controls. The continued exponential growth of WT ESCs at the time when *Nodal* KO1 lines plateaued confirms that this proliferation defect is an artifact of contact inhibition due to over-confluence. Collectively, these suggest that *Nodal* signaling may be indispensable for overcoming the compounded stress of cold-induced pausing and metabolic suppression in mESCs, potentially through a shared metabolic adaptation mechanism uncovered in mTOR inhibition-induced diapause[10]. The highly similar proliferative trajectories shown between the two independent KO clones confirm that the phenotype is *Nodal*-dependent and not due to a clonal artifact. However, the significant phenotypic variability between the *Nodal* KO and WT ESCs under 37°C implies a fundamental variation in their proliferative behaviors. A plausible reason may be the difference in genetic backgrounds between the two cell lines. The *Nodal*-deficient ESCs kindly donated by the E. Robertson lab were derived from a predominantly 129/Sv background, whereas WT E14 mESCs generated in my lab were originally derived from a 129/Ola background[18]. While both belong to the 129 lineage, substrain-specific differences and potential variations from mixed genetic backgrounds in the *Nodal* KO lines may influence cellular phenotypes, which should be considered during the interpretation of my results.

The knockdown experiment suggests a differential requirement for *Cirbp* and *Rbm3* under cold-induced pausing context. Despite CIRBP and RBM3 both being significantly upregulated at 30°C versus 37°C, *Rbm3* depletion generates a strong attenuative effect on cell proliferation compared to *Cirbp* knockdown. This is notable since CIRBP has shown a greater fold-change of upregulation in the proteomic dataset, indicating that its protein expression level may not be fully equivalent to functional importance. The convergence of proliferation trajectories between *Rbm3* KD and double KD conditions further supports that the phenotype in the double KD is

primarily driven by loss of *Rbm3*. However, at the current siRNA concentration adopted, *Rbm3* depletion does not result in a major loss of cell viability in cold-paused mESCs, as cells retain the capacity for limited proliferation. This suggest that *Rbm3* may primarily function to maintain optimal proliferative capacity but not necessarily required for cell survival. In contrast, *Cirbp* knockdown had minimal impact on proliferation under the same conditions, raising the possibility that its role may be cell-type specific or compensated by other regulatory mechanisms in cold-paused mESCs.

Several limitations of the study should be acknowledged. The current *Cirbp* and *Rbm3* knockdown experiment is limited by the transient and incomplete knockdown nature of siRNA-mediated gene silencing, which limits the ability to firmly determine the essentiality of *Cirbp* or *Rbm3*. Residual protein levels may be sufficient to sustain baseline cellular functions and subsequently mask any stronger phenotypes. To address this, ongoing work is now focused on assessing CRISPR-mediated *Rbm3* knockout (KO) mESC lines that are subjected to the same proliferation assay schematic under 37°C and 30°C conditions. The complete KO efficiency was pre-validated using western blot (Figure S1.) for proliferation assay. Complete genetic ablation is expected to demonstrate whether *Rbm3* essentially plays a critical role in ensuring cell survival under cold stress. In parallel, increasing siRNA dosage for a double concentration (60 nM) is currently being tested to enhance knockdown efficiency, particularly for *Cirbp* that shows convergence with NTC under 30°C. If a more potent depletion of *Cirbp* can recapitulate the phenotype observed in *Rbm3* KD conditions, this would explain that the current phenotypic divergence is partially attributable to differences in knockdown efficiency rather than true functional specificity.

In conclusion, my present study identifies *Nodal* and *Rbm3* as potential regulators of mESC proliferation under cold stress conditions. Employing a more complete genetic depletion of *Rbm3* and *Cirbp* will be required to determine the extent to which their expression is necessary for maintaining mESC proliferation and viability during cold-induced pausing.

Methods

Cell culture

Wild-type mouse embryonic stem cells (mESCs) at E14 stage (WT) were maintained on gelatin-coated Nunc 6-well plates in serum LIF (SL) medium made up of DMEM GlutaMAX supplemented with non-essential amino acids (0.1 mM; Gibco, #11140050), fetal bovine serum

(15%; BioTechne, #S12450), penicillin-streptomycin (100 U mL^{-1} and 100 µg mL^{-1}; Gibco, #15140122), 2-mercaptoethanol (0.05 mM; Fisher Scientific, #21985023), and ESGRO LIF (1,000 U mL^{-1}; Sigma, #ESG1107), and were incubated at 37°C with 5% CO_2 and 20% O_2. Cells under 37°C were passaged every two days using 0.05% trypsin-EDTA (Gibco, #25300054). For all experiments, WTE14 mESCs were seeded at 1.0 × 10^5 cells per well as standard density.

Nodal KO proliferation assay

Two independent *Nodal* knockout (KO1 and KO2) mESC lines were kindly provided by lab member Giacomo Furlan who received from E. Robertson lab (Sir William Dunn School of Pathology). Wild-type E14 (WT) and *Nodal* KO mESCs were seeded in technical triplicates on Day 0 (D0). Five hours post-seeding, SL media was refreshed. Plates were sent for imaging immediately after, then maintained at either 37°C or 30°C. Images were taken every 24 hours using the Incucyte SX5 Live-Cell Analysis System, and cell proliferation was quantified using the Adherent Cell-By-Cell Analysis module with user-defined analysis parameters. To avoid over-confluence, WT and *Nodal* KO cells cultured at 37°C were imaged on Day 2 morning to obtain the cell count before splitting. The cells were then replated at a nominal density of 1.0 × 10^5 cells, and imaged 5 hours later to obtain the D2 baseline. Cell numbers for Day 3 were extrapolated by multiplying the 24-hour fold-change (Day 3 count / Day 2 baseline) by the actual Day 2 count. Cell numbers for subsequent days were extrapolated using the same approach. The proliferation curves were generated in GraphPad Prism (v9).

siRNA transfection and proliferation assay

SMARTpool siRNAs targeting *Cirbp* (Horizon Discovery, ON-TARGETplus #L-044409-01-0005) and *Rbm3* (Horizon Discovery, siGENOME #M-041823-01-0005) were purchased (5 nmol). A list of full siRNA sequences is provided in Table 1. The non-targeting siRNA (Horizon Discovery, ON-TARGETplus #D-001810-0X) was available in the lab and used as a negative control. Upon arrival, siRNAs were resuspended in 50 µL nuclease-free water to generate 100 µM stocks. On the day of transfection, 7.5 µL of 100 µM siRNA was diluted in 30 µL of nuclease-free water to generate a 20 µM working stock.

Wild-type E14 mESCs (WT) were seeded in technical duplicates on the evening before transfection in 6-well plates containing 1.5 mL serum LIF medium for 37°C and 30°C. For each temperature condition, two parallel plates were prepared with one well for each siRNA

condition to collect for RNA and protein. To achieve a single knockdown with siRNA concentration at 30 nM in 1.5 mL SL media, 3.5 µL of the diluted siRNA working stock was added to 125 µL Opti-MEM (tube 1) and mixed with 7.5 µL Lipofectamine 3000 in 125 µL Opti-MEM (tube 2). 187.5 uL of the mixture was added into 1.5 mL SL media. For double knockdowns, 1.75 uL *Cirbp* siRNA and 1.75 uL *Rbm3* siRNA were added to each well. Cultures were maintained at 37°C with images taken 5 hours post-transfection. Cells were maintained under either 37°C or 30°C as planned thereafter. Cell proliferation was quantified using the Incucyte SX5 system, with cell count tracked over six days and images taken every 24 hours. 37°C ESCs were passaged every two days, while 30°C cells remained unsplit with SL media changed on the same days.

CRISPR-mediated *Rbm3* knockout

Three independent *Rbm3* knockout mESC lines and one empty-guide control line were previously generated in wild-type E14 mESCs (WT) using the CRISPR-Cas9 system by lab member Ying Yu Ng.

Two distinct guide RNAs (gRNAs) targeting the *Rbm3* locus were designed and cloned into separate pSpCas9(BB)-2A-GFP (PX458-GFP) plasmids (Addgene #48138; gift from F. Zhang). gRNA sequences were shown in Table 2. WT ESCs were transfected with the respective gRNA-containing plasmids or empty PX458-GFP vector using Lipofectamine 3000 (Life Technologies, #L3000015). At 48 hours post-transfection, GFP-positive cells were isolated through fluorescence-activated cell sorting (FACs) and seeded at low density on 96-well plates to retrieve monoclonal lines. As the cell continued expanding, selected clonal lines were validated for the absence of RBM3 protein expression by western blot analysis, following the same schematic described below.

RNA extraction & qRT-PCR

Following 48 hours post-transfection, total RNA was extracted from siRNA-transfected cells using the RNeasy Mini Kit (Qiagen, #74106) according to the manufacturer's instructions, including on-column DNase digestion. RNA concentration and purity were assessed using a Nanodrop instrument (Fisher Scientific, #134005182P5). 200 ng RNA was reverse transcribed using SuperScript VILO cDNA Synthesis Kit (Invitrogen, #11754050) following a step-cycle program: incubate at 25°C for 10 minutes – 42°C for 60 minutes – terminate at 85°C for 5 minutes. cDNA was pre-diluted to 4 ng/uL and mixed with PowerUp SYBR Green Master Mix

(Life Technologies, A25777) as cDNA master mix. Each primer mix contained one pair of quantitative PCR primers spanning exon-exon junctions of *Cirbp* and *Rbm3* mRNA, designed using NCBI Primer-BLAST (Table 3). The cDNA master mix and primer mix were loaded as planned. qRT-PCR was performed on QuantStudio 5 Real-Time PCR System (Thermo Fisher, software v1.5.1) and analyzed with Comparative CT ΔΔCt method for relative expression. The expression levels were normalized to Ubb (housekeeping), with sample values referred to the non-targeting control sample within each temperature condition.

Protein extraction & Western blot

Following 72 hours post-transfection, siRNA-transfected cells were lysed in RIPA buffer containing 150 mM NaCl, 1% IGEPAL NP-40 substitute, 0.5% sodium deoxycholate, 0.1% SDS, and 50 mM Tris-HCl pH 8.0, supplemented with protease and phosphatase inhibitors (Life Technologies, #78442). Lysates were kept on ice for 30°C, followed by a centrifugation for 10 minutes at 20,000 × g. Supernatants were collected to perform a BCA assay to determine protein concentrations. 20 ug protein of each sample was mixed with 6.25 uL non-reducing 4X and nuclease-free water based to achieve a final volume of 25 uL. The prepared sample was denatured at 95°C for 10 minutes, vortexed briefly, and spined down. Proteins were loaded on 4 to 15% Mini-PROTEAN TGX SDS-PAGE gels (Bio-Rad, #4561083) and then transferred to nitrocellulose membranes (Bio-Rad, #1704271). Membranes were cut as planned and blocked in 5% milk prepared in Tris-buffered saline with 0.1% Tween 20 (TBST) for 1 hour, then incubated overnight at 4°C with primary antibodies of target proteins. On the second day, membranes were washed in TBST for 10 minutes, repeating three times. 3 uL HRP-conjugated secondary antibodies was mixed with 30 uL 5% milk to incubate membranes for 1 h at room temperature and then washed again three times. Chemiluminescent signal was developed using ECL reagents (Life Technologies, #32106) or Clarity Max (Bio-Rad, #1705062). Blots were imaged using Bio-Rad Image Lab Software (v6.1).

Tables

Table 1. siRNA sequence targeting *Cirbp* and *Rbm3*

Genes	siRNA sequences
siRNA Cirbp	UGCUACACACAACGAGUAA
	GAGUCAGGGUGGCAGCUAU
	CGGGAGGGUCCUACAGAGA
	CCUCGUGACUAGACCUGUU
siRNA Rbm3	UAGGAGGGCUCAACUUCAA
	UGAGAGCUAUGAAUGGAGA
	AGACUACAGUGGCAGAAGC
	GCGCCAAAUCCGAGUUGAU

siRNA sequences used for the knockdown of *Cirbp* and *Rbm3* are listed. The fours sequences were pooled for each target gene to enhance to reduce off-target effects.

Table 2. Guide RNA sequences targeting *Rbm3* for gene knockout

Guide	Sequence (5'-3')
Guide 1	AGGGAGTGCACGAATTATTG
Guide 2	GGTCCTTGACAACAACCACT

Two single guide RNAs (gRNAs) designed to target the *Rbm3* locus are listed with their corresponding sequences in the 5' to 3' orientation.

Table 3. qPCR primer sequences for *Cirbp* and *Rbm3*

Name	Primer sequence (5'-3')
Cirbp_F1	GAGGACTCAGCTTCGACACC
Cirbp_R1	TCCTTTACCACCACCACTTCG
Cirbp_R2	TGGCTATGAATGGGAAGTCTGT
Cirbp_R2	TCCTCCTCTGGAGAACCCTC
Rbm3_F1	CCTTCTGCCATGTCGTCTGA
Rbm3_R1	TCCTTGACAACAACCACCTCA
Rbm3_F2	GAGCTATGAATGGAGAGTCCCTG
Rbm3_R2	AGACCGCCCATACCCATATC

Designed primer sequences (5'-3') used for *Cirbp* and *Rbm3* qRT-PCR quantification are listed. F stands for forward primer; R stands for reverse primer.

Supplementary data

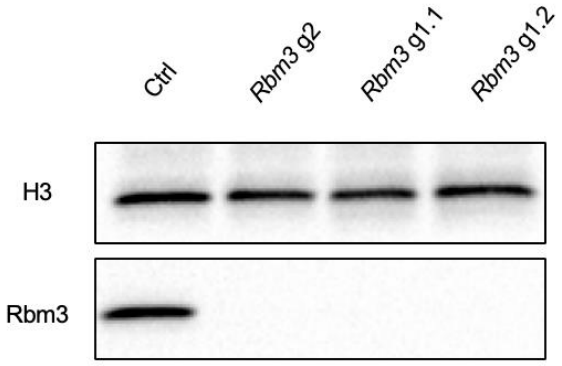


Figure S1. Validation of *Rbm3* knockout lines. Western blot was performed with proteins collected *Rbm3* knockout ESCs on Day 2 under 37°C conditions, with H3 as loading control. The blot confirms the successful knockdown of RBM3 across the two knockout lines targeted with guide 1 (*Rbm3* g1.1 and *Rbm3* g1.2) and one knockout line targeted with guide 2 (*Rbm3* g2). Protein level of the empty-vector transfection control (Ctrl) was retained.

Reference

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